

Retail Sales Analytics Dashboard

Project Report | Microsoft Power BI

1. Project Overview

The **Retail Sales Analytics Dashboard** is an interactive Business Intelligence project developed using Microsoft Power BI. It converts retail transaction data into a visual reporting solution for monitoring sales, orders, customers, products, profit, stores, brands, regions and time-based performance.

The report contains two main analytical pages. The first page provides an overall retail sales view with regional, brand, product, store and city analysis. The second page focuses on time-series performance using previous-year comparison, monthly sales, quarterly channel performance and month-wise regional MTD sales.

2. Project Objectives

- Analyze overall retail sales and profitability.
- Track total orders, customers, products and stores.
- Compare sales across regions and cities.
- Analyze profit contribution by brand.
- Identify top-performing products.
- Compare sales performance across stores.
- Analyze Online and Store channel performance.
- Compare previous-year and current-year sales.
- Analyze monthly and quarterly sales trends.
- Analyze MTD sales by month and region.
- Provide interactive filters for business-focused analysis.

3. Dataset Information

The Power BI project uses a retail sales data model containing a central fact table supported by dimension tables for **Store, Product, Customer and Calendar**. The model enables analysis across geography, products, brands, categories, stores, channels and time.

Data Component	Purpose
Fact Table	Transactional sales data used for sales, orders, profit and time-based measures
Dim Store	StoreID and channel information for store and channel analysis
Dim Product	ProductID, Brand and Category information for product and brand analysis
Dim Customer	Customer, City and Region information for geographic analysis
Dim Calendar	Date, Year, Month, Month Name and Month Number for time-series analysis

4. Tools & Technologies Used

- **Microsoft Power BI:** dashboard development, interactive reporting and visualization.
- **Power Query:** data cleaning and transformation.
- **DAX:** KPI, aggregation and time-intelligence calculations.
- **Data Modeling:** relationships between fact and dimension tables.
- **Power BI Visuals:** KPI cards, bar charts, pie charts, treemaps, line/area charts and tables.

5. Data Cleaning & Preparation

- Imported the retail fact and dimension tables into Power BI.
- Reviewed table structures, column names and data types.
- Used Power Query for data cleaning and transformation.
- Prepared sales, order, customer, product, store and profit fields for analysis.
- Prepared Year, Month, Month Name and Month Number fields for time-based reporting.
- Sorted Month Name by Month Number so months appear in chronological order.
- Created relationships between the fact table and Store, Product, Customer and Calendar dimensions.
- Created DAX measures for Total Sales, Total Orders, Total Customers, Total Products, Total Profit and Total Stores.
- Created time-based measures such as YTD, MTD and Previous Year Sales.
- Formatted monetary values into readable M and K units.
- Added interactive slicers for business analysis.

6. Dashboard Features

- **KPI Cards:** Total Sales, Total Orders, Total Customers, Total Products, Total Profit and Total Stores.
- **Regional Analysis:** Sales by Region.
- **Brand Analysis:** Profit by Brand.
- **Product Analysis:** Top 10 Products by Sales.
- **Store Analysis:** Sales by StoreID and store-level Brand A analysis.
- **City Analysis:** Offline profit and offline sales by city.
- **Time Series:** Previous Year vs Current Year Sales and Sales by Month.
- **Channel Analysis:** Sales by Quarter across Online and Store channels.
- **Regional MTD:** MTD Sales by Month Name and Region.
- **Interactive Slicers:** Channel, StoreID, Brand, Region, City, Year and Month.

7. Key Dashboard Metrics

Metric	Value Shown
Total Sales	49.64M
Total Orders	5K
Total Customers	500
Total Products	200
Total Profit	11.23M
Total Stores	5

8. Dashboard Screenshot – Retail Sales Overview

Dashboard 1 – Retail Sales Overview

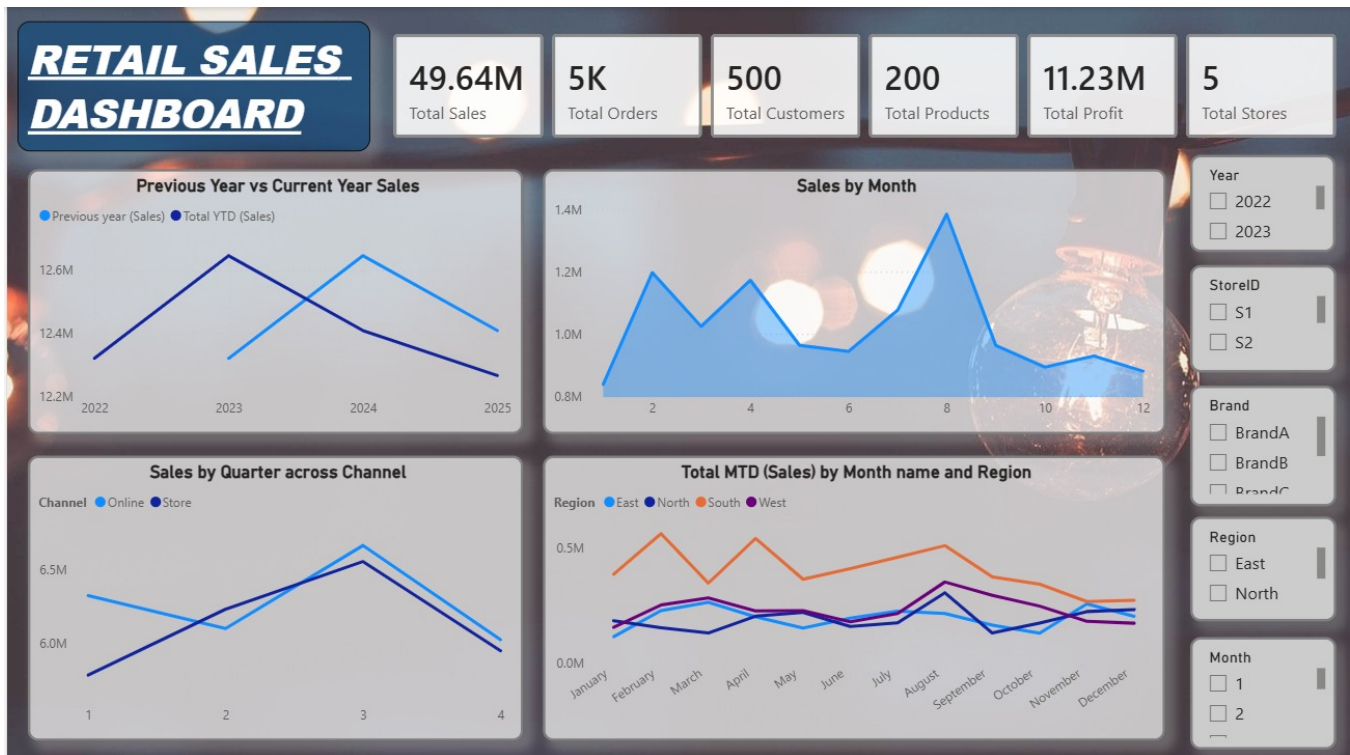


9. Key Insights – Overview

- The dashboard shows **49.64M total sales** and approximately **11.23M total profit**.
- The dataset contains approximately **5K orders, 500 customers, 200 products and 5 stores**.
- The **South** region is the strongest visible region, generating just over 20M in sales.
- West, North and East each contribute around the 9–10M range in the displayed overview.
- Brand-level profit is distributed across Brand A, Brand B, Brand C, Brand D and Brand E, with Brand D showing the largest visible share at about **3M (26.74%)**.
- The Top 10 Products chart shows **P50** as the highest visible product by sales.
- The city table shows Bangalore with the highest displayed offline sales among the listed cities.
- The StoreID treemap provides a quick comparison of Brand A sales across S1 to S5.

10. Dashboard Screenshot – Time Series Analysis

Dashboard 2 – Retail Sales Time Series Analysis



11. Key Insights – Time Series

- The Previous Year vs Current Year Sales visual enables direct comparison of sales performance across the available years.
- The monthly sales chart shows clear fluctuations, with the highest visible monthly sales occurring around **Month 8**, at approximately **1.4M**.
- The quarterly channel chart shows both Online and Store channels peaking around **Quarter 3** in the displayed data.
- The Online channel is slightly higher than the Store channel in Quarter 3, while both decline again in Quarter 4.
- The MTD regional chart shows the **South** region as the strongest visible contributor across most months.
- The time-series page supports filtering by Year, StoreID, Brand, Region and Month, making the analysis interactive.

12. Conclusion

The **Retail Sales Analytics Dashboard** successfully demonstrates how Microsoft Power BI can transform retail transaction data into an interactive Business Intelligence solution. The project combines KPI reporting, regional and brand analysis, product and store analysis, city-level reporting, channel comparison and time-series analysis.

The dashboard provides a clear view of overall sales performance while allowing users to investigate specific regions, brands, stores, products and periods. It demonstrates practical skills in **Power BI, Power Query, DAX, data modeling, time intelligence, data visualization and dashboard design**.

Project Title: Retail Sales Analytics Dashboard | **Technology:** Microsoft Power BI | **Author:** Anjali Patnaik